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1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPI.py
2 The network model is PINNs
3 Layers: ModuleList(
4   (0): Linear(in_features=1, out_features=20, bias=True)
5   (1): Linear(in_features=20, out_features=20, bias=True)
6   (2): Linear(in_features=20, out_features=1, bias=True)
7 )
8 Activation function: Tanh()
9 Total trainable parameters: 481
10 Model on device: cuda:0
11 Start training...
12 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
13 Training: 100%|██████████| 2000/2000 [00:05<00:00, 338.75epoch/s, Loss=1.
  37e-04, LR=1.0e-03, Best=1.37e-04]
14 Training finished. Best loss: 1.06e-04. Model saved to results/best_model.pth
15 Loss history saved to results/loss_history.npy
16 Training time: 6.705195426940918s
17 Relative loss: 0.041039422154426575
18 L2 loss: 0.026113441213965416
19 Parameter containing:
20 tensor([[ -1.2420],
21         [ -0.1518],
22         [  0.8187],
23         [ -1.1588],
24         [ -1.7107],
25         [  0.3478],
26         [ -0.5855],
27         [  1.4668],
28         [  0.9921],
29         [ -0.5645],
30         [ -1.2683],
31         [ -0.5963],
32         [  1.5219],
33         [ -0.0658],
34         [  0.0472],
35         [ -0.5978],
36         [ -0.4336],
37         [ -0.3143],
38         [  1.2709],
39         [ -0.7662]], device='cuda:0', requires_grad=True)
40 Parameter containing:
41 tensor([ 1.0560, -0.1520, -0.2439,  0.8387,  1.4726, -0.4167, -0.1205,  0.1729,
42         -0.1962, -0.5332, -0.6317, -0.2288, -0.4023,  0.6675,  0.9368, -0.3753,
43         -0.1897, -0.4851,  0.1789, -0.6785], device='cuda:0',
44         requires_grad=True)
45 Parameter containing:
46 tensor([[ 3.6261e-01, -2.0151e-01,  4.6561e-02,  3.8839e-01,  3.4095e-01,
47         6.6174e-02,  1.9345e-01, -1.6383e-02, -1.4255e-01,  1.3360e-01,
48         3.2603e-02,  2.1919e-02, -7.0658e-02,  2.1811e-01,  1.1299e-01,
49         -1.0508e-02, -1.2647e-01,  4.7159e-02, -1.2925e-01,  2.0164e-02],
50         [-8.7772e-01, -1.3821e-01,  6.6944e-01, -1.2738e+00, -1.0814e+00,
51         2.4596e-01, -1.8370e-01, -2.5469e-02,  4.2916e-01,  2.3878e-01,

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52	1.5731e-01, -2.4288e-01, -1.5235e-01, -2.1059e-01, -2.1572e-01,
53	2.7513e-02, -1.7997e-01, -1.2509e-01, -2.0388e-01, 1.2282e-01],
54	[2.6832e-01, -1.0232e-01, -2.1828e-01, 4.1258e-01, -2.6619e-02,
55	9.5264e-02, 2.9979e-02, 2.0367e-01, -6.9241e-02, 8.8376e-02,
56	9.6383e-02, 1.1544e-01, -1.1284e-01, 5.8785e-02, 1.1676e-01,
57	-1.9580e-01, 9.8749e-02, 5.5842e-02, -1.5484e-02, -3.0312e-02],
58	[-2.8820e-03, -2.7320e-01, 1.3087e-03, -3.6399e-01, -2.0118e-01,
59	-1.4713e-01, -1.3148e-01, 1.6330e-01, 1.6125e-01, 1.1758e-01,
60	1.4073e-01, -1.9986e-01, -2.2854e-01, 4.5010e-02, -1.2838e-01,
61	1.6069e-01, 1.1464e-01, 1.2467e-03, 6.4125e-02, -1.3658e-01],
62	[1.9588e-01, 3.5069e-02, -4.1752e-01, 6.8262e-01, 4.6899e-01,
63	-2.8550e-01, 2.2811e-01, -1.0151e-01, -4.6248e-01, 1.3988e-01,
64	1.9548e-01, 2.5003e-01, -1.8635e-01, 9.6061e-03, -5.5764e-02,
65	2.0085e-01, -7.1901e-02, -1.2955e-01, 8.6327e-02, 1.2465e-02],
66	[8.2655e-02, 3.5445e-02, -9.0166e-02, -2.9142e-02, 8.4956e-02,
67	2.4512e-03, -2.0047e-02, -8.7016e-02, -2.4553e-01, 7.1741e-02,
68	-5.1855e-02, -1.6675e-01, 1.5175e-01, 9.4643e-02, -6.3126e-03,
69	-1.0202e-01, -1.9713e-02, 1.5245e-01, -8.3004e-02, 1.2614e-01],
70	[-6.4614e-02, 1.2016e-01, -1.6152e-01, -9.5389e-02, -3.2935e-01,
71	1.8735e-01, 6.4068e-02, 1.6373e-01, 4.2075e-02, 1.9281e-01,
72	2.0095e-01, -6.5657e-02, 1.4725e-01, 9.9108e-02, -1.3085e-01,
73	7.3576e-02, 2.0455e-02, 8.0860e-02, 5.2649e-02, -1.0820e-02],
74	[-9.6278e-02, 1.6864e-01, -7.4537e-02, 1.4909e-01, 3.0103e-01,
75	2.4991e-01, 7.0593e-02, -2.9368e-03, -1.1034e-02, 2.1424e-01,
76	1.3042e-01, 3.9428e-02, 2.6377e-01, -3.8377e-02, 1.4503e-01,
77	-1.1212e-01, 7.2216e-03, 2.7178e-01, 2.1694e-01, 8.6793e-02],
78	[-8.2711e-02, -1.4786e-01, -5.5897e-02, 7.8823e-02, -2.3413e-01,
79	5.5583e-02, 1.4453e-01, 1.4378e-01, -8.0710e-02, -2.1997e-01,
80	-4.2505e-02, -1.3278e-02, -1.6100e-01, 2.2251e-01, -5.1713e-02,
81	-2.4786e-02, -1.7362e-01, -2.0513e-02, -1.2546e-01, -1.4854e-01],
82	[3.2326e-01, 1.0772e-01, -5.8734e-01, 3.5997e-01, 3.5021e-01,
83	-1.0596e-01, 1.1668e-01, -7.7014e-03, -2.1973e-01, 1.6310e-01,
84	9.6071e-02, 1.6850e-01, -3.6738e-01, -1.5365e-01, -8.6066e-02,
85	1.8653e-01, 9.1470e-02, 2.2338e-01, 6.3770e-02, 6.5670e-02],
86	[3.6734e-01, 1.3300e-01, -1.9289e-01, 8.9095e-01, 5.0534e-01,
87	-3.8694e-01, 1.0636e-01, 2.2531e-01, -8.3181e-02, 5.1105e-02,
88	-1.5537e-01, 7.3502e-02, -1.5397e-01, 1.0876e-01, 9.2543e-03,
89	8.7386e-02, 2.0296e-01, -5.8099e-02, 8.6788e-02, -5.0934e-02],
90	[1.6065e-02, -2.5256e-02, -4.5559e-03, -3.9885e-02, -7.6769e-02,
91	-7.0908e-02, -1.0770e-01, 1.3807e-01, -2.1936e-02, 2.2410e-01,
92	1.3197e-01, 7.7135e-02, 1.0854e-02, 7.2946e-02, -3.5812e-02,
93	1.0710e-01, -6.8997e-02, 2.7039e-05, -1.3777e-01, -1.9190e-02],
94	[-9.9294e-02, -9.6531e-02, -1.4316e-01, 7.4653e-02, 8.6664e-02,
95	-1.8152e-02, 1.1432e-01, 3.4436e-02, -1.0735e-01, 9.1096e-02,
96	-6.9445e-02, 5.4148e-02, 1.3757e-01, 1.5249e-02, 1.5997e-01,
97	4.4580e-02, 2.5451e-01, -7.6256e-02, 1.7575e-01, -4.3717e-02],
98	[6.7128e-02, -2.6383e-01, 2.0540e-02, 3.8506e-04, 8.0744e-02,
99	-2.2068e-01, -2.3600e-01, -1.6517e-01, 6.4977e-02, -6.2708e-03,
100	-2.0526e-01, -5.0708e-02, 5.5969e-02, -1.4456e-01, -1.0787e-01,
101	1.5631e-01, -1.0550e-01, -1.7077e-01, -3.1720e-02, -2.1790e-01],
102	[-5.1862e-01, -4.5750e-01, 6.5326e-01, -5.9009e-01, -5.1259e-01,
103	4.5678e-01, -6.4989e-01, 5.7853e-01, 8.4133e-01, -5.0913e-01,
104	-3.2407e-01, -9.3858e-02, 6.3623e-01, 2.1015e-01, 3.1136e-02,

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105     -4.2813e-01, -3.3004e-01, -1.2997e-01,  6.0199e-01, -3.8937e-01],
106     [-4.5752e-02, -8.7811e-02,  1.2475e-01, -3.0356e-01, -1.7442e-02,
107     2.9879e-01, -3.9367e-02,  2.4837e-01,  3.2758e-01,  1.9383e-02,
108     -1.0669e-01, -1.5134e-01,  1.8561e-01, -1.2484e-01, -1.9259e-01,
109     -7.0026e-02,  1.8457e-01, -1.8310e-02, -3.7223e-03, -2.2488e-01],
110     [ 1.0003e-01, -1.4238e-01, -1.8057e-01, -2.0898e-01, -8.3800e-02,
111     -6.1005e-02,  1.5685e-01, -1.4754e-01,  1.3536e-02, -2.5043e-01,
112     -8.0228e-02, -2.6306e-01,  4.9290e-02,  2.0233e-01,  2.7575e-01,
113     1.0637e-01,  9.1053e-02, -1.4341e-02, -3.9760e-02,  1.9739e-02],
114     [ 7.6050e-02, -7.9042e-02, -4.5142e-01,  5.6993e-01,  3.5898e-01,
115     -2.4868e-01,  1.7857e-01, -1.6467e-01, -1.1996e-01,  1.7977e-01,
116     2.5660e-02,  1.4348e-01, -2.4511e-01,  4.0740e-02,  4.1759e-02,
117     -6.6304e-02,  1.8092e-01, -2.0147e-01, -9.8960e-02,  2.0843e-01],
118     [-1.6403e-01,  3.2314e-02, -7.9354e-02, -1.4099e-01, -1.2143e-01,
119     1.6320e-02, -1.4678e-01,  7.2460e-02,  1.4632e-01, -1.8992e-01,
120     1.1719e-01,  4.8034e-02, -2.6308e-01,  3.6108e-02, -7.8941e-02,
121     9.6509e-02, -2.3339e-01, -2.0628e-01, -1.3136e-01, -1.7080e-01],
122     [ 2.7973e-01, -8.4631e-03,  2.2371e-02,  7.2309e-01,  1.1464e-01,
123     -2.5358e-01,  1.3985e-01, -8.1493e-02, -1.8860e-01, -1.6269e-02,
124     -1.8153e-01,  6.8481e-02, -2.1733e-01,  2.1730e-01,  1.7464e-01,
125     2.2496e-01, -5.6676e-02, -1.6603e-01, -3.8970e-03,  5.7789e-02]],
126     device='cuda:0', requires_grad=True)
127 Parameter containing:
128 tensor([ 0.2444, -0.2349, -0.0327, -0.0786, -0.1718,  0.1265,  0.1682, -0.0195,
129         0.1860,  0.1790,  0.1489, -0.2217, -0.2316,  0.1873, -0.1810, -0.0532,
130         0.0735, -0.1560,  0.0764,  0.2213], device='cuda:0',
131         requires_grad=True)
132 Parameter containing:
133 tensor([[ 0.2491, -0.1392,  0.1509, -0.1441, -0.1516, -0.0057, -0.0854,  0.2852,
134         -0.1581, -0.2085,  0.4909,  0.1073,  0.0504, -0.1889,  0.3898,  0.2310,
135         -0.2198, -0.1144, -0.3738,  0.1355]], device='cuda:0',
136         requires_grad=True)
137 Parameter containing:
138 tensor([-0.0858], device='cuda:0', requires_grad=True)
139 =====
140 Layer (type:depth-idx)          Output Shape          Param #
141 =====
142 PINNModel                      [1, 1]                --
143 |—ModuleList: 1-5              --                      (recursive)
144 |   |—0.weight                  |—20
145 |   |—0.bias                    |—20
146 |   |—1.weight                  |—400
147 |   |—1.bias                    |—20
148 |   |—2.weight                  |—20
149 |   |—2.bias                    |—1
150 |   |—Linear: 2-1              [1, 20]              40
151 |   |   |—weight                |—20
152 |   |   |—bias                  |—20
153 |—Tanh: 1-2                    [1, 20]              --
154 |—ModuleList: 1-5              --                      (recursive)
155 |   |—0.weight                  |—20

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156 |      | 0.bias | 20
157 |      | 1.weight | 400
158 |      | 1.bias | 20
159 |      | 2.weight | 20
160 |      | 2.bias | 1
161 |      | Linear: 2-2 | [1, 20] | 420
162 |      | | weight | 400
163 |      | | bias | 20
164 | Tanh: 1-4 | [1, 20] | --
165 | ModuleList: 1-5 | -- | (recursive)
166 |      | 0.weight | 20
167 |      | 0.bias | 20
168 |      | 1.weight | 400
169 |      | 1.bias | 20
170 |      | 2.weight | 20
171 |      | 2.bias | 1
172 |      | Linear: 2-3 | [1, 1] | 21
173 |      | | weight | 20
174 |      | | bias | 1
175 | =====
176 |      |
177 |      |
178 |      |
179 |      |
180 | =====
181 |      |
182 |      |
183 |      |
184 |      |
185 | =====
186 |      |
187 |      |
188 |      |

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